

Institute of Actuaries of India

Subject CT1 – Financial Mathematics

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) Assume that the retail price of the toy is 100. The cash price is 70 or one can pay 75 in 6 months' time. Hence the effective rate of discount d is

$$70/75 = (1-d)^{0.5}, \text{ Where } d = 0.1289 \text{ or } 12.89\% \quad [1]$$

ii)

- $d^{(12)} = 12 \times [1 - (1 - d)^{1/12}] = 13.72\%$,
- $i = \frac{d}{(1-d)} = 14.79\%$, and $i^{(2)} = 2 \times [(1 + i)^{1/2} - 1] = 14.28\%$
- $(1+i) = e^\delta$ hence $\delta = 0.1380$ [3]

[4 Marks]

Solution 2:

i)
$$\begin{aligned} V(t) &= \exp\left[-\int_0^t d(s) ds\right] \\ &= \exp\left[-\int_0^t a e^{-bs} ds\right] \\ &= \exp\left[\frac{a}{b}(e^{-bt} - 1)\right] \end{aligned} \quad [1]$$

ii) Since $\delta(0) = 0.01$, $a = 0.01$ and $\delta(10)/\delta(0) = e^{-10b} = 0.5$
 $b = 0.069315$, [2]
 Hence the present value of the payment is
 $1000 [v(1) + v(2) + v(3) + v(4)] = 3910.84$

iii) solving the equation $1000 (e^{-\delta} + e^{-2\delta} + e^{-3\delta} + e^{-4\delta}) = 3910.84$
 $\Rightarrow \delta = 0.00904$ [1]

[4 Marks]

Solution 3:

Let K_t and S_t denote the forward & stock price of the contract at time t respectively.

Then, $K_0 = S_0 e^{0.0875}$ at time $t = 0$

$K_{3/4} = S_{3/4} e^{0.078 \times 1/4}$ at time $t = 0.75$ (or 9 months)

At the beginning of contract, value will be zero and at 9 months its is $V_{3/4} = (K_{3/4} - K_0) e^{-0.078 \times 1/4}$

For the contract to remain neutral, its value should also be zero.

Hence,

$$V_{3/4} = (K_{3/4} - K_0) e^{-0.078 \times 1/4} = 0 \text{ Or } K_{3/4} = K_0$$

$$S_{3/4} e^{0.078 \times 1/4} = S_0 e^{0.0875}$$

$$S_{3/4} / S_0 = e^{(0.0875 - 0.025 \times 0.078)} = 1.070365$$

Hence shares of ABC should rise to INR 1070.37 to keep the contract value neutral against change in environment conditions.

[5 Marks]

Solution 4:

All amounts in crores

PV of outgo = $10 \cdot \bar{a}_2$ at 10%

$$\bar{a}_2 = \frac{i}{\delta} \cdot a_2 = 1.049206 \times 1.7355 = 1.82090$$

=> PV of outgo = 18.209

Income is: 8 at the end of year 3, 7½ at the end of year 4, 3 at the end of year 13, 2 at the end of year 14 and 1 at the end of year 15.

PV of income = $v^2 \times 0.5 \times (Da)_{10} + v^2 \times 3 \times a_{10} + v^{12} \times (Da)_3$ at 10%

$$(Da)_{10} = \frac{10 - a_{10}}{i} = \frac{10 - 6.1446}{0.10} = 38.5543$$

$$((Da)_3 = \frac{3 - 2.4869}{0.10} = 5.1310$$

=> PV of income = 15.9314 + 15.2345 + 1.6349 = 32.8008

=> NPV of project = 32.8008 - 18.2090,

= 14.592

[5 Marks]

Solution 5:

The expected accumulated value can be calculated as:

$$2,400 E(S_5) = 3,600$$

$$2,400 (1 + j)^5 = 3,600$$

⇒ $j = 8.45\%$, where j is the mean rate of interest

The variance of accumulated value can be calculated as:

$$\text{var}(2,400 S_5) = 2,400^2 \text{var}(S_5) = 50^2$$

$$\text{var}(S_5) = 50^2 / 2400^2 = [(1 + j)^2 + s^2]^5 - (1 + j)^{10}$$

$$50^2 / 2400^2 = [1.0845^2 + s^2]^5 - 1.0845^{10}$$

Solving for s , will give:

$s = 0.006735$, where s is variance of interest rate

As the interest rate follows *Gamma* (α, λ) distribution; we can determine its parameters:

Let i_k be the interest rate in year k .

$$j = E(i_k) = \alpha \mid \lambda = 0.0845$$

$$s^2 = \text{var}(i_k) = \alpha \mid \lambda^2 = 0.006735^2$$

Solving for parameters:

$$\alpha = 157.3$$

$$\lambda = 1862.23$$

[7 Marks]

Solution 6:

i) i_1 , Spot rates at time 1 = $0.09 - 1/200 + 1/500 = 8.7\%$

i_2 , Spot rates at time 2 = $0.09 - 4/200 + 2/500 = 7.4\%$

i_3 , Spot rates at time 3 = $0.09 - 9/200 + 3/500 = 5.1\%$

Equation to determine par yield (Y_p) is:

$$P = P / (1 + i_3)^3 + Y_p [1/(1 + i_1) + 1/(1 + i_2)^2 + 1/(1 + i_3)^3], \text{ where } P \rightarrow \text{Par value}$$

$$100 = 100 / (1 + i_3)^3 + Y_p [1/(1 + i_1) + 1/(1 + i_2)^2 + 1/(1 + i_3)^3]$$

$$100 = 100 / 1.051^3 + Y_p [1/1.087 + 1/1.074^2 + 1/1.051^3]$$

$$13.86 = Y_p * 2.648$$

$$Y_p = 5.23\% \quad [5]$$

ii) Coupon bias is the difference between the par yield rate and the spot rate

Or

Defined as the difference between the yield of a zero coupon bond and yield of a coupon bearing bond.

In this case, it will be the difference of $= Y_p - i_3 = 0.13\%$

[2]

[7 Marks]

Solution 7:

Working in 1000's

i) $X = 24 \times 360 \times a_6^{(12)} \text{ at } 6\% = 8135.8 \quad [2]$

ii) Balance at 1st January 2011

$$Y = 8135.8 (1.06) - [12 \times 360 \times s_1^{(12)}] 6\% = 4186.4$$

Consider quarterly periods commencing 1st January with interest at 1% per quarter

$$\text{The balance as at 1st April 2011 is } Z = Y (1.01) - [3 \times 360 s_1^{(3)}] \text{ at } 1\% = 3144.7$$

The balance as at 1st October 2011 is

$$Z (1.01)^2 - [3 \times 460 s_2^{(3)}] \text{ at } 1\% = 424.9$$

Even adding interest for a month, this would still be less than 460 so there will be insufficient balance for the rent at end of October 2011 [4]

iii) Let balance at 1st April 2011 be increased to z' sufficient to provide rent until the end of 2011

$$Z' (1.01)^3 - [3 \times 460 s_3^{(3)}] \text{ at } 1\% = 0$$

$$Z' = 4071.9$$

$$\text{Extra money needed} = Z' - Z = 927.2 \text{ i.e. } 927,200$$

[3]

[9 Marks]

Solution 8:

i) Time weighted rate of return is given by:

$$(1+i)^2 = \frac{500}{460} \times \frac{(550+50)}{(500+40)} \times \frac{x}{550}$$

$$(1+i)^2 = 1.44$$

$$\Rightarrow X = (1.44 \times 550) / (1.086957 \times 1.1111)$$

$$\Rightarrow X = 655.78$$

[2]

ii) Money weighted rate of return is given by the equation:

$$460 (1+i)^2 + 40(1+i)^{7/4} - 50(1+i) = 655.78$$

$$\text{For } i = 20\%, \text{ LHS} = 657.4335$$

$$\text{For } i = 19\%, \text{ LHS} = 646.1394$$

$$\text{So, MWRR} = 19.85\% \text{ p.a.}$$

[2]

iii) The effective rate of return for the year 2015 is given by solving the equation of value:

$$460 (1+i) + 40 (1+i)^{\frac{3}{4}} = 600$$

$$\Rightarrow i = 20.44\%$$

The effective rate of return for the year 2016

$$(1+i) = \frac{655.78}{550} = 1.1923$$

So the linked rate of return is

$$(1+i)^2 = 1.1923 \times 1.2044$$

$$\text{Hence LIRR} = 19.83\%$$

[2]

- iv) When the sub-intervals chosen for calculating the Linked internal rate of return coincide with the intervals between net cash flows being received then the Linked internal rate of return will be identical to the time weighted rate of return. [1]

v)

- ⇒ Time Weighted Rate of Return (TWRR) is a better measure of fund manager's performance
- ⇒ Money-weighted rate of return (MWRR) is sensitive to the amounts and timings of the net cash flows, but fund manager does not have any control on these.
- ⇒ Time-weighted rate of return (TWRR) is calculated using the "growth factors" reflecting the change in fund value between the times of consecutive cash flows. Thus TWRR is independent of amount and timing of cash-flows.
- ⇒ Hence Time-weighted rate of return (TWRR) is a better measure of fund manager's performance [2]

[9 Marks]

Solution 9:

- i) The discounted payback period of an investment project is the first time when the net present Value of the cash flows from the project is positive [1]

- ii) a) Let us work in units of 1000, to find the DPP
 $-800(1+i)^t - 50(1+i)^{t-1} + 100 \bar{s}_{t-2}$ at 7% = 0

At t= 17 the net present value is -75

At t = 18 the net present value 22.85

Hence By interpolation t = 17.767 [3]

- b) The accumulated profit after 22 years is found by accumulating the income after DPP has elapsed at 6% p.a.

$$100 \bar{s}_{4.233} \text{ at } 6\% = 480$$

The accumulated amount is INR. 480,000 [2]

- iii) The businessman may accumulate his rental income for each year at 6%,

$$100\bar{s}_1 = 102.971 \text{ at the end of the year}$$

The DPP is thus the smallest integer t such that

$$-800(i+i)^t - 50(i+i)^{t-1} + 102.971s_{t-2} \text{ at } 7\% \geq 0$$

By trial and error t = 18 years

The balance in the businessman's account just after the transaction at time 18 years is

$$-800(i + i)^{18} - 50(i + i)^{17} + 102.971 s_{16} \text{ at } 7\% = 9.7714$$

Hence the accumulated amount after 22 years is

$$9.7714 (i + i)^4 + 100\bar{s}_4 \text{ at } 6\%$$

$$= 462.79$$

[6]

Hence the accumulated amount after 22 years is INR. 462,790

[12 Marks]

Solution 10:

- i) Immunization of Portfolio – Portfolio of securities where some investments are assets (positive cash flow) like loans and others are a liability (negative cash flow) like zero coupon bonds. Immunization is an investment strategy to minimize the interest rate change risk of a portfolio by adjusting the investment duration. [2]
- ii) Consider the loan issue size of INR X and INR Y given by bank for tenure of 6 and 15 years respectively.

In order for the portfolio to be immunized, it should satisfy the Redington's three conditions, which could be used to determine the issue size of X & Y.

The first condition requires present value of assets and liabilities to be the same. Hence,
 $V_L = 5 v^8 + 10 v^{11} @ 8\% = \text{INR } 6.99 \text{ m (payment for zero coupon bonds)}$

$$V_A = X v^6 + Y v^{15} @ 8\% \text{ (recovery of loan by customers)}$$

$$V_A = V_L \text{ giving us our first equation}$$

$$0.63 X + 0.32 Y = 6.99 \quad \text{--- (A)}$$

By Redington's second condition, volatility of asset and liability should be equal. Hence,

Derivative of liability portfolio:

$$V_L' = 5 * 8 * v^9 + 10 * 11 * v^{12} @ 8\% = 63.69$$

Derivative of asset portfolio:

$$V_A' = X * 6 * v^7 + Y * 15 * v^{16} @ 8\% = 3.5 X + 4.38 Y$$

$$V_A' / V_A = V_L' / V_L$$

$$V_A' = V_L' \quad [:: V_A = V_L]$$

$$3.5 X + 4.38 Y = 63.69 \quad \text{--- (B)}$$

Solving equation A & B, we get

$$X = \text{INR } 6.37 \text{ m}$$

$$Y = \text{INR } 9.45 \text{ m}$$

[6]

- iii) We now check the portfolio for Redington's third equation. The convexity of assets and Liabilities needs to be determined:

$$V_L'' = 5 * 8 * 9 * v^{10} + 10 * 11 * 12 * v^{13} @ 8\% = 652.11$$

$$V_A'' = X * 6 * 7 * v^8 + Y * 15 * 16 * v^{17} @ 8\% = 757.5$$

Since $V_A = V_L$, convexity of assets is greater than that of liability hence the 3rd condition of Redington is satisfied and thus the portfolio is immunized.

[3]

[11 Marks]

Solution 11:

Working in units of 100

- i) Let $x/12$ be the monthly repayment. Solving the equation

$$X a_{25}^{(12)} = 9880 \text{ at } 7\%$$

$$X = 821.76$$

$$X/12 = 821.76/12 = 68.48$$

Hence the monthly repayment is INR 6848

[2]

- ii) The loan outstanding immediately after the repayment on 10th march 2029

$$X a_{34/3}^{(12)} = 648,600 \text{ at } 7\%$$

[2]

- iii) The loan outstanding just after the repayment on 10th September 2026 is made

$$X a_{83/6}^{(12)} \text{ at } 7\%$$

The loan outstanding just after the repayment on 10th October 2026 is made

$$X a_{13.75}^{(12)} \text{ at } 7\%$$

The capital repaid on 10th October 2026 is thus

$$X \left[a_{83/6}^{(12)} - a_{13.75}^{(12)} \right] \text{ at } 7\%$$

$$= X \left[\frac{v^{13.75} - v^{83/6}}{i^{(2)}} \right] \text{ at } 7\%$$

$$= 26.86$$

[4]

- iv) (i) The capital repayment continued in these 12 instalments is

$$X \left[a_{22/3}^{(12)} - a_{19/3}^{(12)} \right] \text{ at } 7\%$$

$$= 516.20$$

- (ii) The total interest in these 12 instalments is

$$X - 516.20 = 305.56$$

[4]

[12 Marks]

Solution 12:

First Step will be to determine GRY available on Government Bonds.

Since the current price of bond is lower than the redemption price, capital gain will be applicable. GRY on Government bond can be determined using the following equation:

$$P_G = R_G / (1 + i_G)^{10} - (R_G - P_G) * t_c / (1 + i_G)^{10}$$

Where,

$P_G \rightarrow$ Price of Government Bond

$R_G \rightarrow$ Redemption price of Government Bond

$t_c \rightarrow$ applicable capital gain tax

$i_G \rightarrow$ GRY on Government Bond

$$\Rightarrow 500 = 1,000 / (1 + i_G)^{10} - (1000 - 500) * 10\% / (1 + i_G)^{10}$$

Solving for i_G will give;

$$i_G = 6.63\% \text{ p.a.}$$

Assuming interest rates are flat for the tenure of debentures; investors (Corporate or Retail) will need GRY of at least 6.63% p.a. from the issue. Hence the price should be determined basis this rate as GRY which will be the maximum, the company can set for its issue.

Coupons on debentures are payable on a quarterly basis, so:

$$i_G^{(4)} @ 6.63\% = 4 \times (1.0663^{1/4} - 1) = 6.47\%$$

This rate is higher than the coupon rate of debentures hence there will be a capital gain for both the set of investors.

To determine maximum price of issue for retail investors, the equation will be:

$$P_R = R_R / (1 + i_G)^{10} - (R_R - P_R) * t_c / (1 + i_G)^{10} + D \times a_{10}^{(4)} \times (1 - t_i)$$

where, $P_R \rightarrow$ Price of Debenture for retail investors

$R_R \rightarrow$ Redemption price of Debenture for retail investors

$t_i \rightarrow$ applicable income tax rate

$D \rightarrow$ Dividend rate on debenture

$$P_R = 1000 / 1.0663^{10} - (1000 - P_R) * 10\% / 1.0663^{10} + 60 \times 7.32 \times (1 - 15\%)$$

$$P_R = 1000 * 0.5263 - (1000 - P_R) * 10\% * 0.5263 + 373.32$$

$$P_R = 899.62 - (1000 - P_R) * 10\% * 0.5263$$

$$P_R * .94737 = 899.62 - 52.63$$

$$P_R = 894.04$$

To determine maximum price of issue for corporate investors, the equation will be:

$$P_C = R_C / (1 + i_G)^{10} - (R_C - P_C) * t_C / (1 + i_G)^{10} + D * a_{10}^{(4)} * (1 - t_{iC})$$

where, $P_C \rightarrow$ Price of Debenture for Corporate investors

$R_C \rightarrow$ Redemption price of Debenture for Corporate investors

$t_{iC} \rightarrow$ applicable income tax rate

$$P_C = 105,000 / 1.0663^{10} - (105,000 - P_C) * 10\% / 1.0663^{10} + 6000 * 7.32 * (1 - 20\%)$$

$$P_C = 105,000 * 0.5263 - (105,000 - P_C) * 10\% * 0.5263 + 35,136$$

$$P_C = 90,397.5 - (105,000 - P_C) * 10\% * 0.5263$$

$$P_C * .94737 = 84,871.35$$

$$P_C = 89,586.27$$

Hence, the maximum issue price from company should be INR 894.04 per nominal INR 1,000 for retail investors. Similarly, the issue price for corporate investors should be INR 89,586.27 per nominal INR 1lac. Any price lower than it will give investors earn better yield than Government bond.

[15 Marks]
